PART II : CHEMISTRY

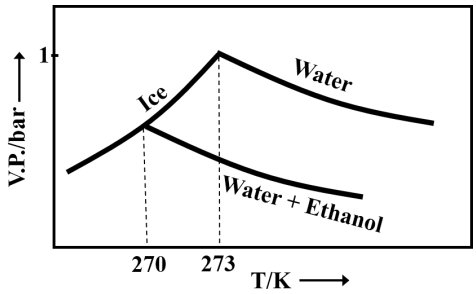
SECTION 1 (Maximum Marks: 21)

- This section contains **SEVEN** questions
- Each question has **FOUR** options [A], [B], [C] and [D]. **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS
- For each question, marks will be awarded in one of the following categories:

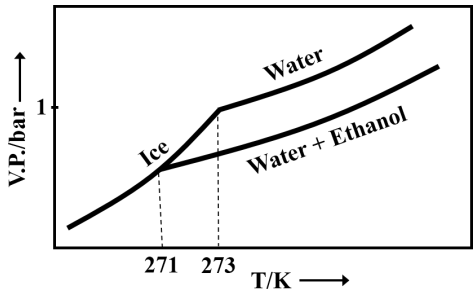
Full Marks	: +3	If only the bubble corresponding to the correct option is darkened
Zero Marks	: 0	If none of the bubbles is darkened
Negative Marks:	-1	In all other cases

Q.19 Pure water freezes at 273 K and 1 bar. The addition of 34.5 g of ethanol to 500 g of water changes the freezing point of the solution. Use the freezing point depression constant of water as 2 K kg mol^{-1}. The figures shown below represent plots of vapour pressure (V.P.) versus temperature (T). [molecular weight of ethanol is 46 g mol^{-1}]
Among the following, the option representing change in the freezing point is

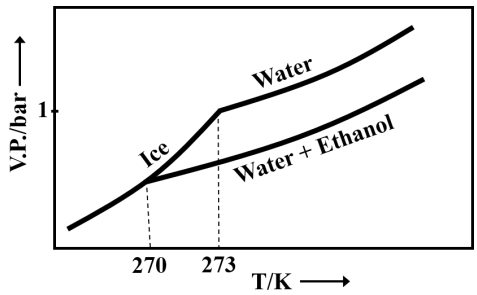
[A]



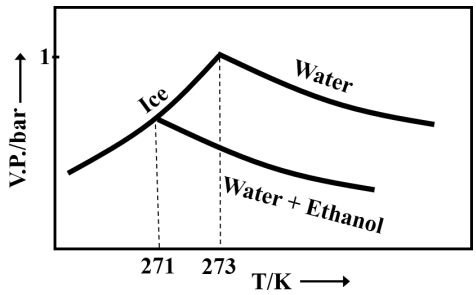
[B]



[C]



[D]



Answer for the above question

Ans for Q.19: (C)

Q.20 For the following cell,

$$Zn(s)\,|\,ZnSO_4(aq)\,\|\,CuSO_4(aq)\,|\,Cu(s)$$

when the concentration of Zn^{2+} is 10 times the concentration of Cu^{2+}, the expression for ΔG (in J mol^{-1}) is

[F is Faraday constant; R is gas constant; T is temperature; $E^o(cell) = 1.1\,\text{V}$]

[A] 1.1F [B] 2.303RT − 2.2F

[C] 2.303RT + 1.1F [D] −2.2F

Q.21 The standard state Gibbs free energies of formation of C(graphite) and C(diamond) at T = 298 K are

$$\Delta_f G^o[\text{C(graphite)}] = 0 \text{ kJ mol}^{-1}$$

$$\Delta_f G^o[\text{C(diamond)}] = 2.9 \text{ kJ mol}^{-1}.$$

The standard state means that the pressure should be 1 bar, and substance should be pure at a given temperature. The conversion of graphite [C(graphite)] to diamond [C(diamond)] reduces its volume by 2×10^{-6} m^3 mol^{-1}. If C(graphite) is converted to C(diamond) isothermally at T = 298 K, the pressure at which C(graphite) is in equilibrium with C(diamond), is

[Useful information: 1 J = 1 kg m^2 s^{-2}; 1 Pa = 1 kg m^{-1} s^{-2}; 1 bar = 10^5 Pa]

[A] 14501 bar [B] 58001 bar [C] 1450 bar [D] 29001 bar

Q.22 Which of the following combination will produce H_2 gas?

[A] Fe metal and conc. HNO_3

[B] Cu metal and conc. HNO_3

[C] Zn metal and NaOH(aq)

[D] Au metal and NaCN(aq) in the presence of air

Space for rough work

Answers for the above questions

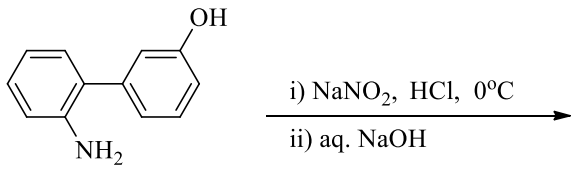
Ans for Q.20: (B)	Ans for Q.21: (A)	Ans for Q.22: (C)

Q.23 The order of the oxidation state of the phosphorus atom in H_3PO_2, H_3PO_4, H_3PO_3, and $H_4P_2O_6$ is

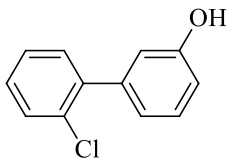
[A] $H_3PO_3 > H_3PO_2 > H_3PO_4 > H_4P_2O_6$ [B] $H_3PO_4 > H_3PO_2 > H_3PO_3 > H_4P_2O_6$

[C] $H_3PO_4 > H_4P_2O_6 > H_3PO_3 > H_3PO_2$ [D] $H_3PO_2 > H_3PO_3 > H_4P_2O_6 > H_3PO_4$

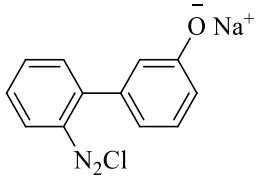
Q.24 The major product of the following reaction is



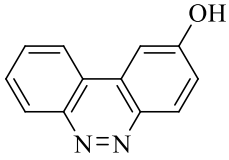
[A]



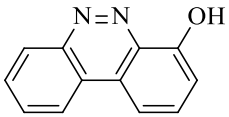
[B]



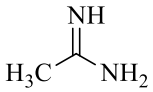
[C]



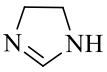
[D]



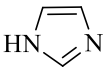
Q.25 The order of basicity among the following compounds is



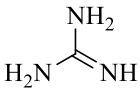
I



II



III



IV

[A] II > I > IV > III [B] IV > II > III > I

[C] IV > I > II > III [D] I > IV > III > II

Space for rough work

Answers for the above questions

Ans for Q.23: (C)	Ans for Q.24: (C)	Ans for Q.25: (C)

SECTION 2 (Maximum Marks: 28)

- This section contains **SEVEN** questions
- Each question has **FOUR** options [A], [B], [C] and [D]. **ONE OR MORE THAN ONE** of these four options is(are) correct
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- For each question, marks will be awarded in one of the following categories:

Full Marks	: +4	If only the bubble(s) corresponding to all the correct option(s) is(are) darkened
Partial Marks	: +1	For darkening a bubble corresponding **to each correct option**, provided NO incorrect option is darkened
Zero Marks	: 0	If none of the bubbles is darkened
Negative Marks	: -2	In all other cases

- For example, if [A], [C] and [D] are all the correct options for a question, darkening all these three will get +4 marks; darkening only [A] and [D] will get +2 marks; and darkening [A] and [B] will get -2 marks, as a wrong option is also darkened

Q.26 The correct statement(s) about surface properties is(are)

[A] Adsorption is accompanied by decrease in enthalpy and decrease in entropy of the system

[B] The critical temperatures of ethane and nitrogen are 563 K and 126 K, respectively. The adsorption of ethane will be more than that of nitrogen on same amount of activated charcoal at a given temperature

[C] Cloud is an emulsion type of colloid in which liquid is dispersed phase and gas is dispersion medium

[D] Brownian motion of colloidal particles does not depend on the size of the particles but depends on viscosity of the solution

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Space for rough work

Answer for the above question

Ans for Q.26: (A) and (B)

Q.27 For a reaction taking place in a container in equilibrium with its surroundings, the effect of temperature on its equilibrium constant K in terms of change in entropy is described by

[A] With increase in temperature, the value of K for exothermic reaction decreases because the entropy change of the system is positive

[B] With increase in temperature, the value of K for endothermic reaction increases because unfavourable change in entropy of the surroundings decreases

[C] With increase in temperature, the value of K for endothermic reaction increases because the entropy change of the system is negative

[D] With increase in temperature, the value of K for exothermic reaction decreases because favourable change in entropy of the surroundings decreases

Q.28 In a bimolecular reaction, the steric factor P was experimentally determined to be 4.5. The correct option(s) among the following is(are)

[A] The activation energy of the reaction is unaffected by the value of the steric factor

[B] Experimentally determined value of frequency factor is higher than that predicted by Arrhenius equation

[C] Since P = 4.5, the reaction will not proceed unless an effective catalyst is used

[D] The value of frequency factor predicted by Arrhenius equation is higher than that determined experimentally

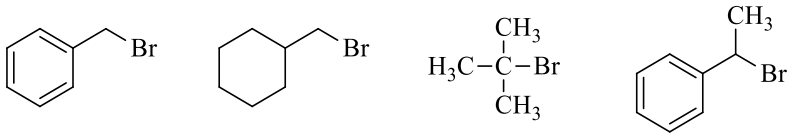
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Space for rough work

Answers for the above questions

Ans for Q.27: (B) and (D)	Ans for Q.28: (A) and (B)

Q.29 For the following compounds, the correct statement(s) with respect to nucleophilic substitution reactions is(are)



I **II** **III** **IV**

[A] **I** and **III** follow S_N1 mechanism

[B] **I** and **II** follow S_N2 mechanism

[C] Compound **IV** undergoes inversion of configuration

[D] The order of reactivity for **I**, **III** and **IV** is: **IV** > **I** > **III**

Q.30 Among the following, the correct statement(s) is(are)

[A] $Al(CH_3)_3$ has the three-centre two-electron bonds in its dimeric structure

[B] BH_3 has the three-centre two-electron bonds in its dimeric structure

[C] $AlCl_3$ has the three-centre two-electron bonds in its dimeric structure

[D] The Lewis acidity of BCl_3 is greater than that of $AlCl_3$

Q.31 The option(s) with only amphoteric oxides is(are)

[A] Cr_2O_3, BeO, SnO, SnO_2 [B] Cr_2O_3, CrO, SnO, PbO

[C] NO, B_2O_3, PbO, SnO_2 [D] ZnO, Al_2O_3, PbO, PbO_2

Space for rough work

Answers for the above questions

Ans for Q.29: {(A), (B), and (C)} or {(A), (B), (C), and (D)}

Ans for Q.30: (A), (B), and (D)

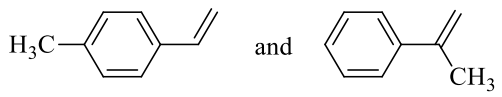
Ans for Q.31: (A) and (D)

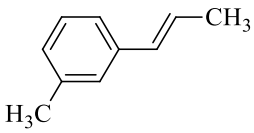
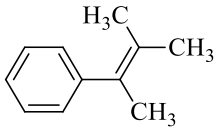
Q.32 Compounds **P** and **R** upon ozonolysis produce **Q** and **S**, respectively. The molecular formula of **Q** and **S** is C_8H_8O. **Q** undergoes Cannizzaro reaction but not haloform reaction, whereas **S** undergoes haloform reaction but not Cannizzaro reaction.

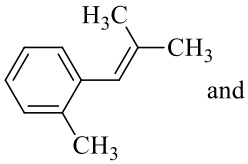
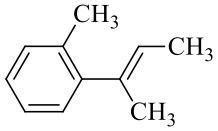
(i) **P** $\xrightarrow[\text{ii) } Zn/H_2O]{\text{i) } O_3/CH_2Cl_2}$ **Q** (C_8H_8O)

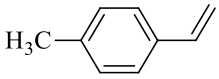
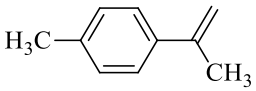
(ii) **R** $\xrightarrow[\text{ii) } Zn/H_2O]{\text{i) } O_3/CH_2Cl_2}$ **S** (C_8H_8O)

The option(s) with suitable combination of **P** and **R**, respectively, is(are)

[A]  and

[B]  and 

[C]  and 

[D]  and 

Space for rough work

Answer for the above question

Ans for Q.32: (A) and (B)

SECTION 3 (Maximum Marks: 12)

- This section contains **TWO** paragraphs
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options [A], [B], [C], and [D]. **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS
- For each question, marks will be awarded in one of the following categories:

Full Marks : +3 If only the bubble corresponding to the correct option is darkened
Zero Marks : 0 In all other cases

PARAGRAPH 1

Upon heating $KClO_3$ in the presence of catalytic amount of MnO_2, a gas **W** is formed. Excess amount of **W** reacts with white phosphorus to give **X**. The reaction of **X** with pure HNO_3 gives **Y** and **Z**.

Q.33 **W** and **X** are, respectively

[A] O_3 and P_4O_6

[B] O_2 and P_4O_6

[C] O_2 and P_4O_{10}

[D] O_3 and P_4O_{10}

Q.34 **Y** and **Z** are, respectively

[A] N_2O_3 and H_3PO_4

[B] N_2O_5 and HPO_3

[C] N_2O_4 and HPO_3

[D] N_2O_4 and H_3PO_3

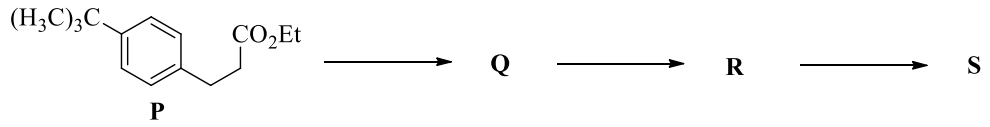
Space for rough work

Answers for the above questions

Ans for Q.33: (C)	Ans for Q.34: (B)

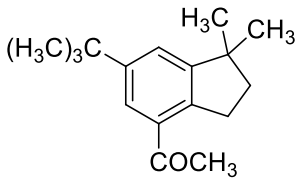
PARAGRAPH 2

The reaction of compound **P** with CH_3MgBr (excess) in $(C_2H_5)_2O$ followed by addition of H_2O gives **Q**. The compound **Q** on treatment with H_2SO_4 at 0°C gives **R**. The reaction of **R** with CH_3COCl in the presence of anhydrous $AlCl_3$ in CH_2Cl_2 followed by treatment with H_2O produces compound **S**. [Et in compound **P** is ethyl group]

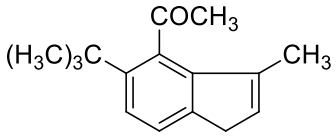


Q.35 The product **S** is

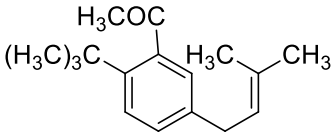
[A]



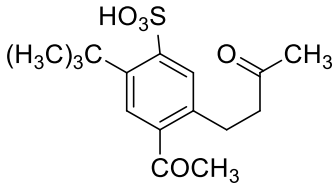
[B]



[C]



[D]



Q.36 The reactions, **Q** to **R** and **R** to **S**, are

[A] Dehydration and Friedel-Crafts acylation

[B] Aromatic sulfonation and Friedel-Crafts acylation

[C] Friedel-Crafts alkylation, dehydration and Friedel-Crafts acylation

[D] Friedel-Crafts alkylation and Friedel-Crafts acylation

END OF PART II : CHEMISTRY

Space for rough work

Answers for the above questions

Ans for Q.35: (A)	Ans for Q.36: (D)